

ADHAM HEGAZY

FULL STACK DEVELOPER

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PROFESSIONAL SUMMARY

Full stack developer with a Bachelor of Science in Computer Science and a strong foundation in full-stack development, specializing in scalable web and mobile applications using React.js, Node.js, and Python. Experienced in designing clean architectures following SOLID principles and best practices. Skilled in integrating AI/ML models including predictive analytics and deep learning using TensorFlow and Keras. Capable of building end-to-end systems with a focus on performance, scalability, and reliability.

EDUCATION

Bachelor of Science in Computer Science — AAST

Sep 2021 – Jul 2025

Arab Academy for Science, Technology & Maritime Transport (AAST)

- GPA: 3.4 / 4.0 — Very Good with Honors

PROFESSIONAL EXPERIENCE

Teaching Assistant — College of Engineering and Technology, AAST

Sep 2026 – Present

- Teach and mentor undergraduate students across Computer Vision, Natural Language Processing, and Digital Logic Design courses, translating dense technical theory into clear, practical explanations.
- Design and deliver hands-on lab sessions and coding exercises in Python (OpenCV, NLP libraries) and digital logic simulation tools, reinforcing programming fundamentals, algorithmic thinking, and system design concepts directly applicable to full-stack development.
- Grade assignments, projects, and exams, and hold regular office hours to debug student code and clarify core CS concepts including data structures, OOP, and problem decomposition.
- Build supplementary course materials and example projects, strengthening communication, technical writing, and cross-topic skills in AI/ML (CV, NLP) that carry directly into building intelligent full-stack applications.

Full Stack Developer — Accountants on Air

Jul 2026 – Aug 2026

- Built ShieldOps, a cybersecurity incident-triage assistant for SOC analysts, using Next.js 14 (App Router), TypeScript, and Tailwind CSS.
- Designed a resilient AI orchestration layer with the Vercel AI SDK, integrating Groq and Google Gemini with configurable primary/fallback providers for incident-response generation.
- Enforced strict request/response validation with Zod and deterministic triage logic, and hardened the service around security-first defaults: server-side-only API keys, no automatic containment/remediation, and stack-trace-free error responses.
- Covered core logic and API routes with a Vitest test suite and documented architecture, API contracts, and known limitations for maintainability.

Mobile Developer Intern — Megatrust

Aug 2023 – Sep 2023

- Built a real-time, WebSocket-based location-tracking layer in Flutter/Dart for a live taxi-hailing app, giving riders and drivers continuously updated ETAs during active trips.
- Integrated payment and geolocation APIs into production, then added response caching and request batching to cut down on redundant network calls and speed up screen loads.
- Shipped 2 production-ready cross-platform apps (iOS + Android) from a single Flutter/Dart codebase, avoiding the cost of building and maintaining separate native codebases.
- Refactored core modules around SOLID principles and worked peer code review into every two-week Agile sprint, catching integration issues before release rather than after.

PROJECTS

Themis — AI-Powered HR Management System

React.js, Python, TensorFlow, Keras, PHP, MongoDB

- Trained an employee-behavior prediction model in TensorFlow/Keras on historical HR data, using a documented train/test split to validate performance before deployment.
- Designed a backend architecture built to support 10,000+ concurrent users with real-time alerting, using MongoDB and a Node.js/PHP service layer.
- Automated candidate shortlisting workflows, replacing a fully manual recruitment screening process and freeing up HR team time for higher-value review.

Promobot — Retail Robot Fleet Platform

React, TypeScript, Node.js, Express, Vite, JWT, REST APIs

- Built a React + TypeScript frontend with protected routes and lazy loading, serving separate, purpose-built dashboards for admin and retail users.
- Secured all REST API traffic by implementing JWT authentication and role-based access control across a Node.js/Express backend.
- Built an end-to-end campaign-creation and approval workflow, letting fleet operators deploy a product promotion to a specific robot without engineering involvement.

Deep Convolutional GAN for Face Generation

Python, TensorFlow, Keras

- Trained a DCGAN on 200,000+ images to generate high-quality synthetic faces, applying targeted data augmentation to improve generalization and managing the full model lifecycle from training through evaluation.

GoShop — E-Commerce Marketplace

React.js, Node.js, REST APIs

- Built reusable React components for product listings and cart management, reducing duplicated UI code across new features and improving overall usability.

TECHNICAL SKILLS

Core CS: Data Structures & Algorithms, OOP, Design Patterns, SOLID Principles, System Design, Clean Code

Languages: Python, JavaScript, TypeScript, Java, C++, C#, C, Dart

Web: React.js, Node.js, REST APIs, HTML5, CSS3, PHP

Mobile: Flutter, React Native

Data: MongoDB, MySQL, PostgreSQL

AI/ML: TensorFlow, Keras, scikit-learn, Deep Learning, LLM API integration (OpenAI, Anthropic)

Tools: Git, Agile/Scrum, Code Review, CI basics

CERTIFICATIONS

- React Basics — Meta
- AI in Applications — Accountants on Air
- Flutter & Dart Mobile Developer — Megatrust
- Digital Transformation & Data Literacy — SAS
- AI Career Essentials — ALX Arabia

LANGUAGES

Arabic: Native • English: Fluent • French: Intermediate